The Paper Plane Challenge – Longest Flight Duration

Technical report – Project X Selection Task (RC Plane, Problem 2) Name: Harshita Mishra · Roll No.: B26036 ·

1. Objective, Constraints and Assumptions

Objective: build a paper airplane that stays airborne for the *longest duration* (not the greatest distance).

Constraints followed: one standard A4 sheet (210 × 297 mm); folding and cutting only; no tape, glue, paper clips or external propulsion; manual launch only; tests indoors in still air.

Assumptions: [A4 sheet of 80 gsm, mass ≈ 5 g]; air density ≈ 1.2 kg/m³; no wind or airconditioning draft; the same person launches every trial from about the same height and angle.

2. Part A – Design

2.1 Construction steps

1. Take one A4 sheet in portrait orientation. Fold it in half lengthwise, crease firmly, and unfold to mark the centreline.
2. Fold the two top corners in so the top edges meet the centreline, forming a pointed nose.
3. Fold the new angled edges in to the centreline again, leaving 2.5 cm from below. This makes a narrow, double-layer nose that moves the centre of gravity forward without adding weight.
4. Fold the small nose tip back over the folded flaps to lock them in place (no tape or glue is allowed).
5. Fold the whole plane in half along the centreline, with the folded flaps on the outside, to form the body.
6. Fold each wing down from the body, leaving a wing width of 8.5 cm, so the wings are broad and swept back in a delta shape.
7. Open the wings flat with a slight upward angle (small dihedral), and fold up the outer wing tips to make small winglets.
8. Adjust the trailing edges slightly upward to trim the plane, so it glides flat instead of diving or stalling.

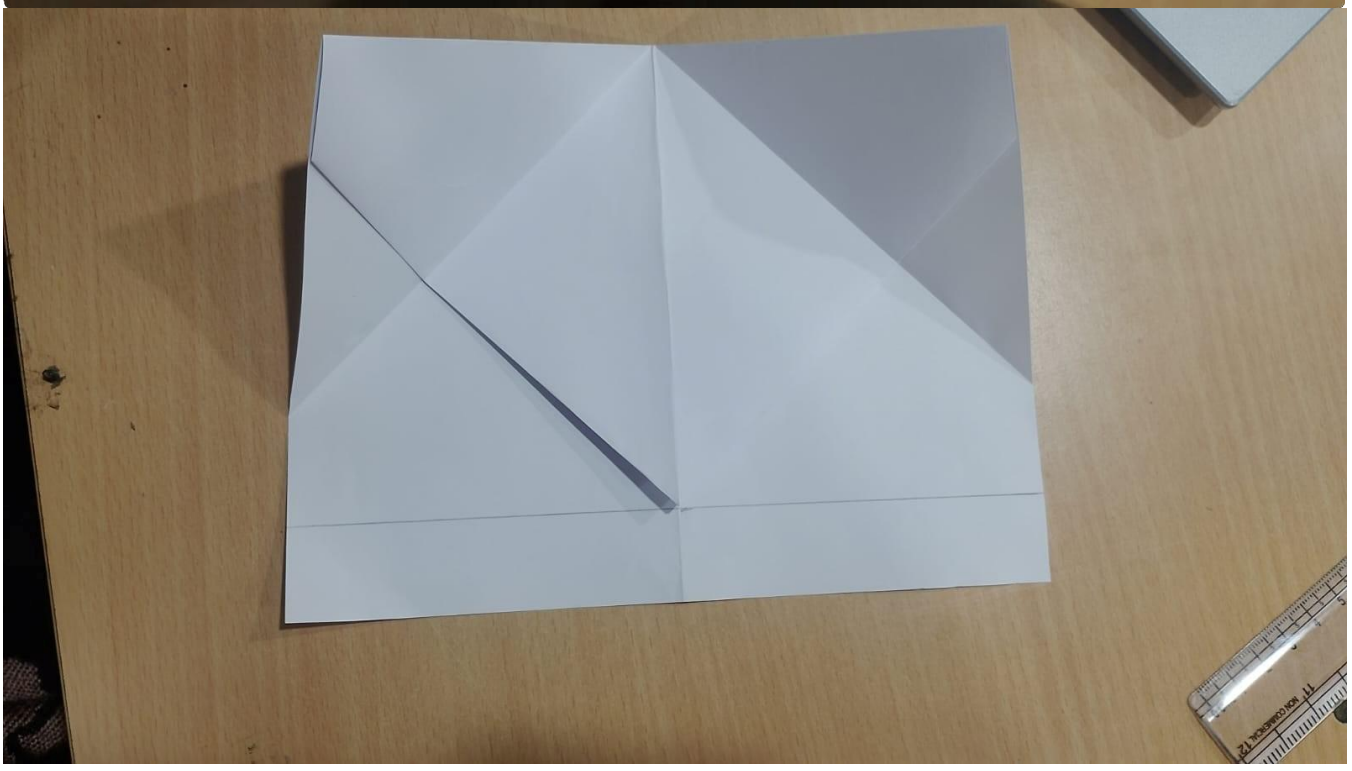
Construction photos, in order from the first fold (photo 1) to the finished plane (photo 9):

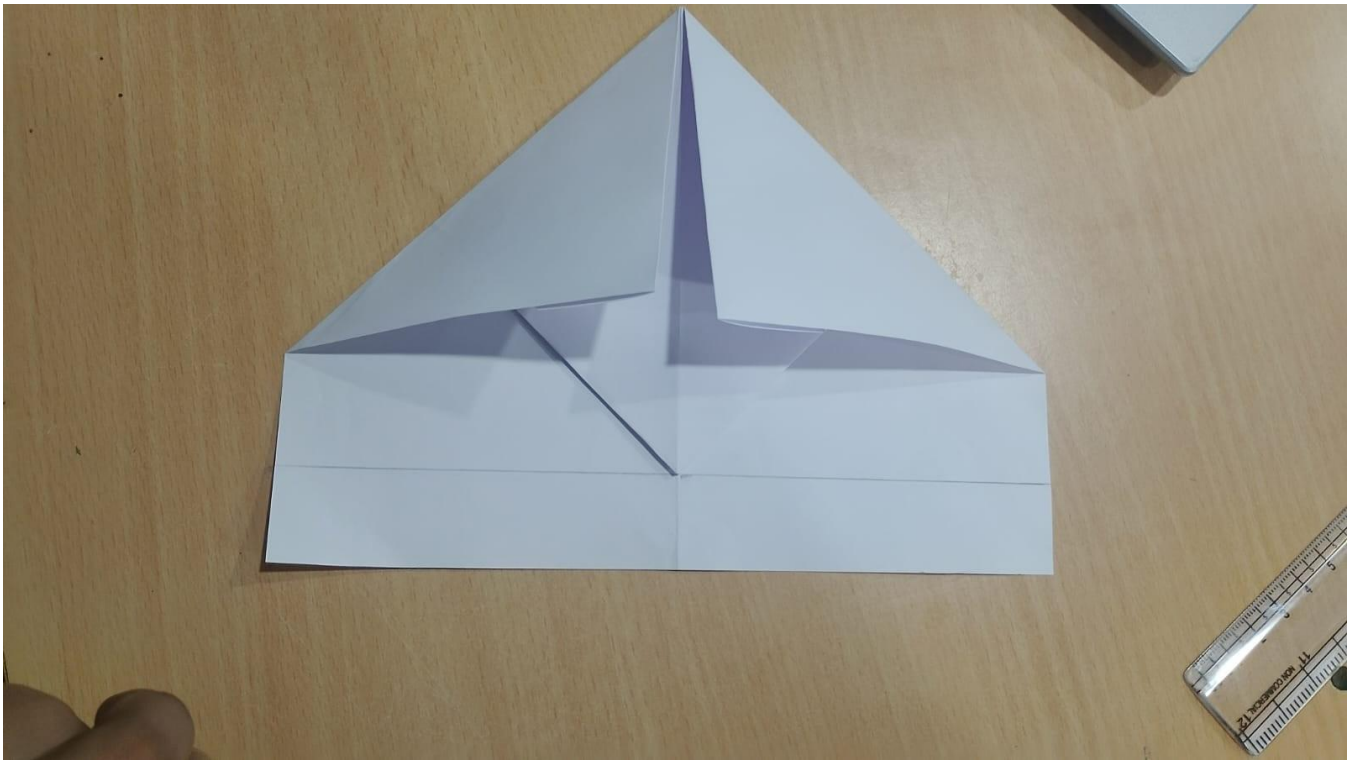




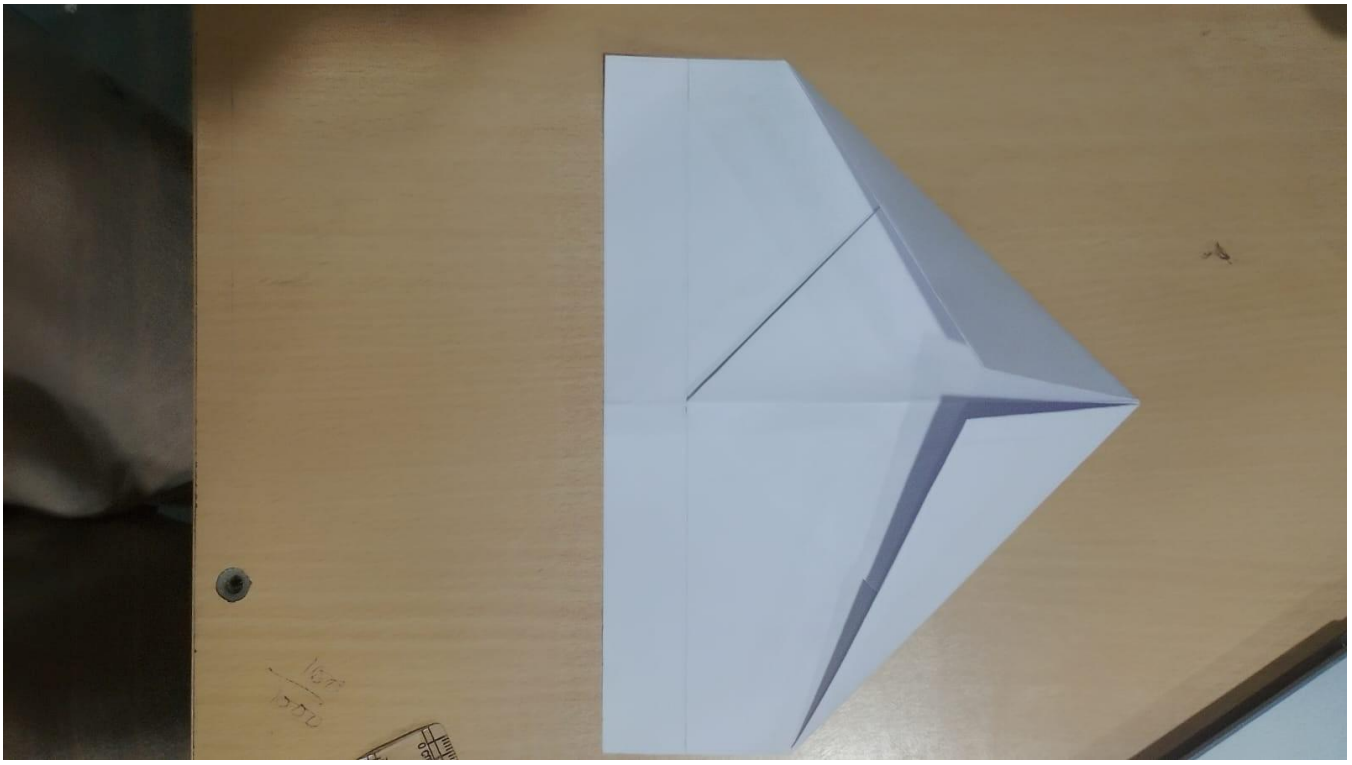






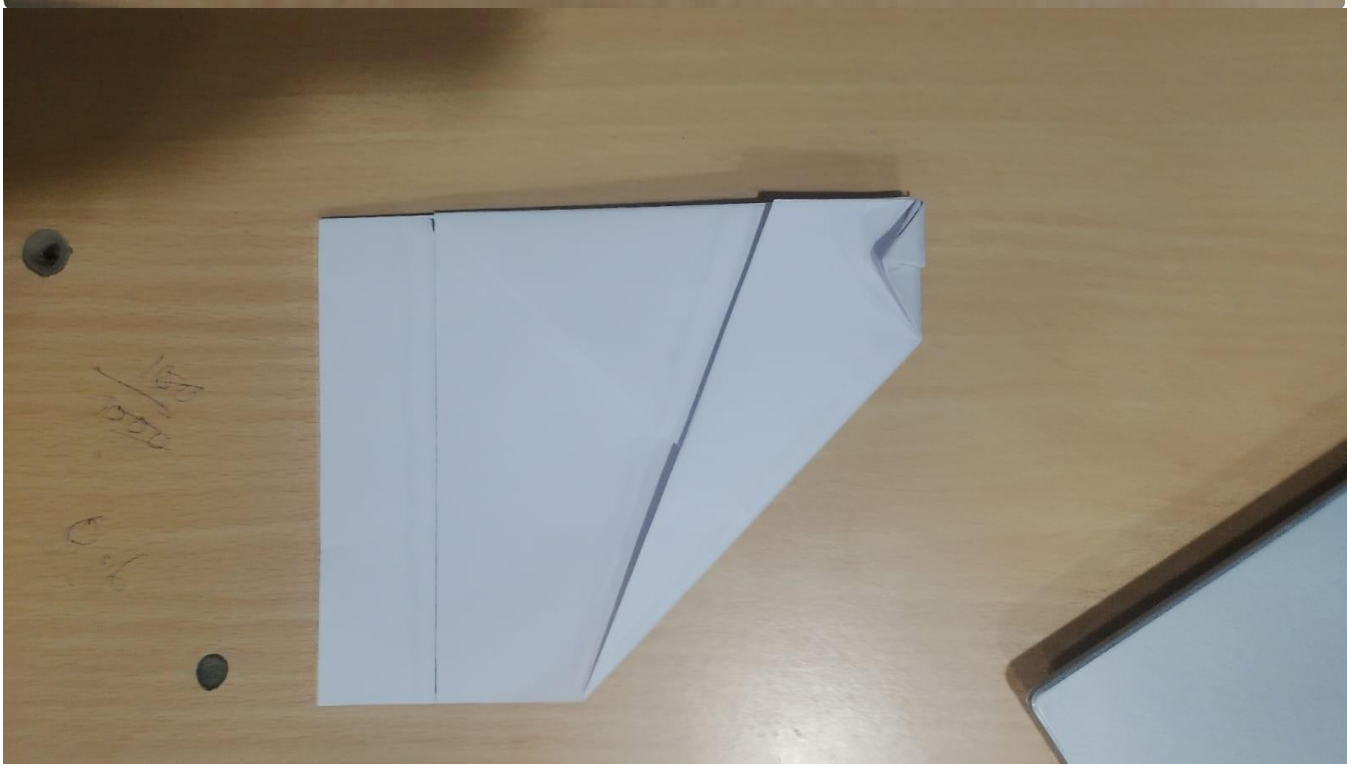


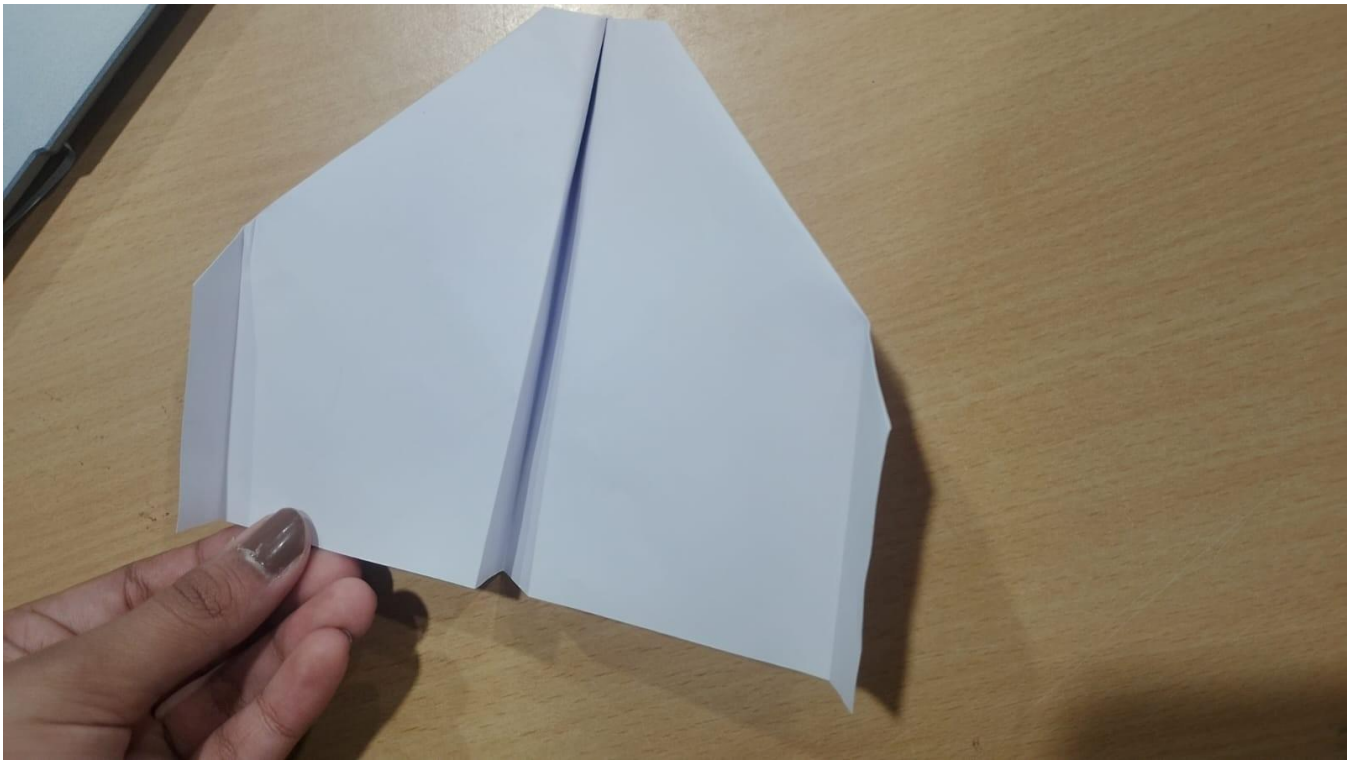












2.2 Design choices and dimensions

Parameter	Value	Reason for choice
Overall length	[11.7cm]	Long body gives directional stability and room to place the CG forward
Wingspan	[17cm]	Larger span gives lower induced drag and lower sink rate
Wing area	[0.003315 m²]	Large area keeps wing loading low, so the plane flies slowly
Wing shape	Broad, swept-back (delta-style) wing with upturned winglets	Straight wing: higher lift; slight sweep: better stability
Dihedral	[~5°]	Self-corrects rolling, keeps the glide steady
Winglets / tip folds	[Yes, 1cm]	Reduce wingtip vortex losses and add roll stability
Elevator trim	[small upward bend]	Fine-tunes pitch so the plane glides rather than dives or stalls
Mass	[~5g]	Fixed by one A4 sheet – no weight can be added
Wing loading (W/S)	[1.48N/m²]	Low wing loading → low sink speed and longer flight
CG position	[[~28]% of wing chord from the leading edge]	Slightly ahead of the aerodynamic centre for stable flight

The design is a **low-wing-loading glider** rather than a fast dart, because the aim is time in the air, not distance.

3. Part B – Aerodynamics

3.1 How the airplane generates lift

As the plane moves forward, the wing, set at a small angle of attack, deflects the air downward. By Newton's third law the air pushes the wing upward. At the same time the airflow over the upper surface speeds up and the pressure there drops, while the pressure below is higher; this pressure

difference acts as lift. A paper wing is flat, so lift comes mainly from the angle of attack (and from any slight curvature in the folds), not from airfoil camber. Lift is approximately $L = ½ \rho V^2 S C_L$.

3.2 Forces during launch and glide

Force	Launch phase	Glide phase
Lift	Increases rapidly with speed and can exceed weight, giving a climb	Equals the component of weight perpendicular to the flight path
Weight	Constant (≈ [0.05] N)	Constant; the component along the path drives the glide
Drag	High because speed is high	Opposes motion; equals the component of weight along the flight path
Thrust	Only the thrown impulse from the hand; no sustained thrust	None – the plane is an unpowered glider

With no sustained propulsion, the launch only adds kinetic energy (and some height). After that, the plane gradually trades height for forward speed to overcome drag. In a steady glide, lift balances the weight component and drag balances the along-path component of weight, so the glide angle γ satisfies $\tan \gamma = D/L = 1/(L/D)$.

3.3 Wing area, wing loading, centre of gravity and stability

- **Wing area (S):** a larger area produces the same lift at a lower speed.
- **Wing loading (W/S):** the weight per unit wing area. The required flight speed is $V = \sqrt{(2W / \rho S C_L)}$, so lower W/S means lower flight speed and a lower sink rate, which gives longer flight time.
- **Centre of gravity (CG):** it must be ahead of the aerodynamic centre (about 25% of the chord). A CG too far forward makes the plane nose-heavy and it dives; too far back makes it stall and tumble. Folding extra layers at the nose moves the CG forward without adding weight.
- **Stability:** longitudinal stability comes from the CG being ahead of the aerodynamic centre and from the tail/elevator; directional stability comes from the long body and keel; roll stability comes from dihedral and winglets.

3.4 Why this design should stay airborne longer than a conventional dart

A conventional dart has a small wing area and a heavy nose, so it must fly fast to produce enough lift. It covers distance quickly and sinks fast. My design has a larger wing area and lower wing loading, so it flies slower and stays in the air longer. Its sink rate is lower, and the dihedral and trim keep the glide steady instead of diving.

Minimum sink rate vs. maximum glide distance:

	Minimum sink rate	Maximum glide distance
What it means	Slowest vertical descent speed	Greatest horizontal distance per unit height lost
Occurs at	The speed that minimises power required, (C_L^1.5 / C_D) maximum	The speed where L/D is maximum
Optimises	Time in the air	Range
Relative speed	Lower speed	Higher speed
Relevant to this task?	**Yes – this is what we want**	No – that would be the distance challenge

Sink rate is $V_s = V \sin \gamma \approx \sqrt{(2W / \rho S)} \cdot (C_D / C_L^{1.5})$. A plane with low W/S and a high $C_L^{1.5}/C_D$ therefore stays up longest.

4. Part C – Experiment

4.1 Test conditions

Location: [indoor hall / room with no draft]. Launch height: 1.45 m (same each time). Launch angle: [slightly upward, ~10°] and a gentle launch speed. Timing: [stopwatch on phone], timed from release until touching the ground. Same launcher and same plane for every trial. Room conditions: [no fans, doors closed].

4.2 Flight trial results (final design)

Trial	Launch height (m)	Launch angle	Flight time (s)	Notes
1	1.45	[~10° up]	4.80	Smooth, long glide; fresh folds
2	1.45	[~10° up]	4.50	Good glide; slight drift to the right
Trial	Launch height (m)	Launch angle	Flight time (s)	Notes
3	1.45	[~10° up]	4.00	Nose started to dip; wing tips slightly bent after landing
4	1.45	[~10° up]	3.80	Glide steeper; creases looser

5	1.45	[~10° up]	3.43	Shorter glide; plane visibly deformed
Mean			**4.11**	(4.80 + 4.50 + 4.00 + 3.80 + 3.43) / 5 = 20.53 / 5

Best flight: 4.80 s (trial 1). Worst flight: 3.43 s (trial 5). Sample standard deviation ≈ 0.55 s.

4.3 Results and discussion

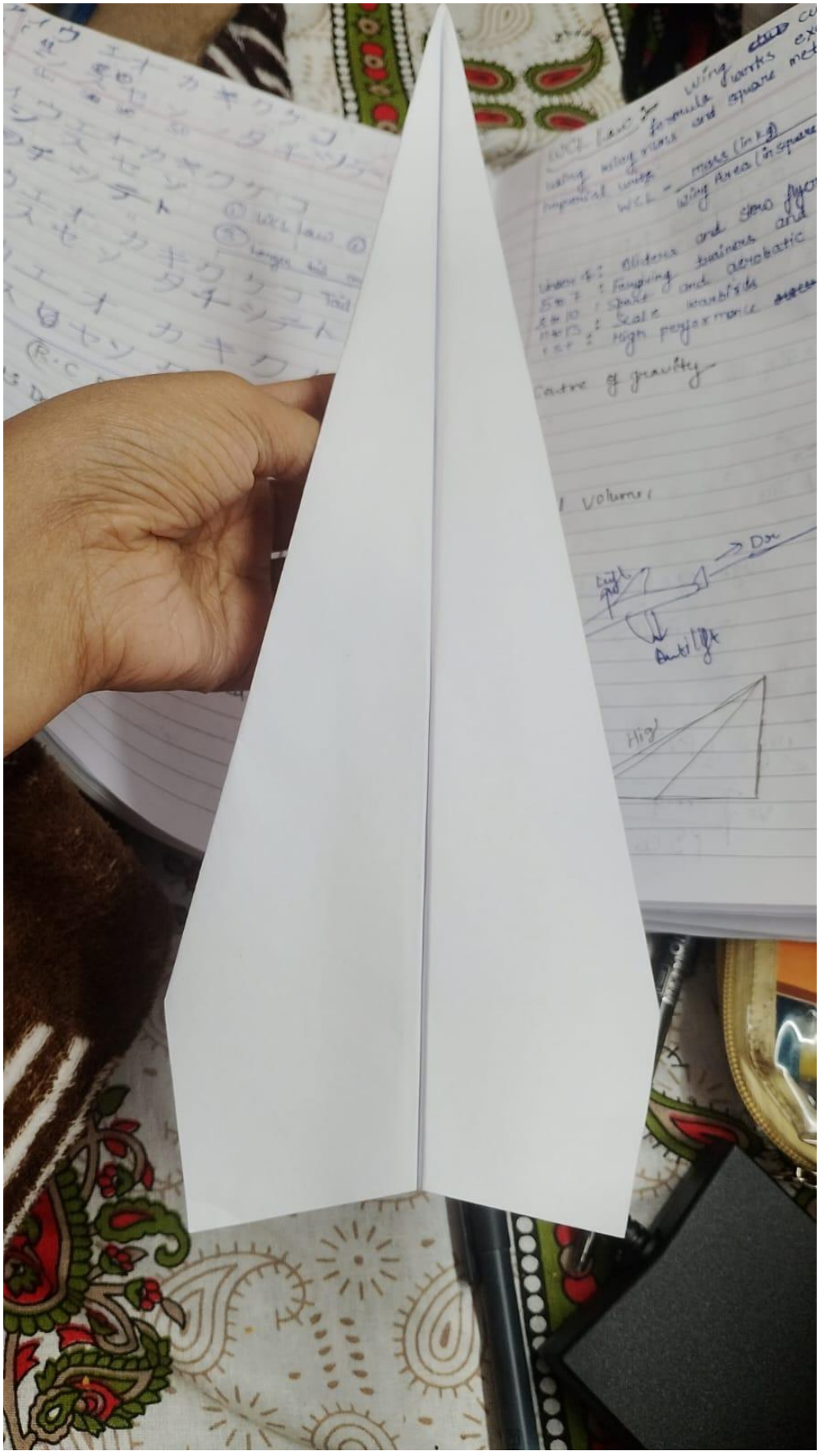
The final design averaged 4.11 s over five trials, with a best of 4.80 s and a worst of 3.43 s (standard deviation ≈ 0.55 s). The flight time fell steadily from trial to trial, about 29% from the first trial to the fifth, instead of scattering randomly. This points to a systematic cause rather than chance: [the folds and wing creases loosened with repeated handling, the wings or nose were slightly bent by landings, or the launch became less consistent as I repeated it – edit to the cause you actually observed]. Re-creasing the folds and checking the wing shape between trials would be needed for fully repeatable results. [Describe how the plane flew: smooth glide, stall, dive or spiral.]

4.4 Suggested modifications and how to test them

Modification	Expected effect	Variable changed	Variables kept constant	Test method
[Increase dihedral from 5° to 10°]	[More roll stability, steadier glide, slightly lower lift]	Dihedral angle	Wing area, mass, CG, launch height/angle, same launcher and room	5 trials with the modified plane, compare the mean time with the baseline
[Move CG forward by folding extra layers at the nose]	[Reduces stalling, but too much causes diving]	CG position	Wing shape, dihedral, launch conditions	5 trials per CG setting, record mean time and flight behaviour
Modification	Expected effect	Variable changed	Variables kept constant	Test method
[Wider wingspan with smaller chord (higher aspect ratio)]	[Lower induced drag, longer glide]	Aspect ratio	Total sheet, mass, launch conditions	5 trials, compare means

Only one modification is changed at a time so any difference in mean flight time can be attributed to that change.

5. Bonus – Second Design Comparison



Design	Mean flight time (s)	Trials	Best (s)	Worst (s)

Design 1 (final)	4.11	5	4.80	3.43
Design 2	2.50	5	3.00	1.60

Design 2 trials (launch height 1.45 m):

Trial	Launch height (m)	Launch angle	Flight time (s)
1	1.45	[~10° up]	2.55
2	1.45	[~10° up]	3.00
3	1.45	[~10° up]	2.70
4	1.45	[~10° up]	1.60
5	1.45	[~10° up]	2.63
Mean			**2.50**

Design 2 standard deviation ≈ 0.53 s (Design 1: ≈ 0.55 s).

Conclusion: Design 1 stayed in the air about 64% longer on average (4.11 s vs. 2.50 s), and even Design 1's worst flight (3.43 s) beat Design 2's best flight (3.00 s). Design 2 have high wing cube loading which leads to maximum sink rate and minimum time of flight and also its tail size doesn’t balance its nose so it just dives faster into the ground .

6. Conclusion

The plane was designed as a low-wing-loading glider to maximise time in the air by minimising sink rate. The mean flight time over five trials was 4.11 s (range 3.43–4.80 s). Wing area, wing loading, CG position and dihedral were the main factors controlling the result, and the downward trend across trials shows that keeping the folds crisp matters too. Further improvement can be tested by changing one design variable at a time.